



In the name of Allah

Application of natural mineral biochar from Kerman mountains as a supplement in livestock and poultry feed production

is used as an effective supplement in the production of animal and poultry feed. will be

These rich nutrients have beneficial properties for the health of livestock and poultry



By Mr. Engineer Mohammad Majidi Kouhbanani



The effect of natural mineral biochar of Kerman mountains on improving the quality of livestock feed and poultry parameters

1

Reduce nutrient loss

using _ The natural mineral biochar of Kohbanan Kerman can reduce the loss of nutrients in animal and poultry feed.

2

Increasing livestock health

The natural mineral biochar of Kerman mountain can guarantee the improvement of the quality and health of livestock.

3

Improve growth performance

Research shows that using From Biochar The natural mineral of Kerman mountains can improve their growth performance in feeding livestock and poultry



Reducing the wastage of nutrients in livestock and poultry feed using natural mineral biochar from Kerman's mountains

1

Environmental protection

The use of natural mineral biochar from Kerman mountains can lead to the reduction of environmental reactions related to the loss of nutrients.

2

Disease control

By absorbing mineral elements and various indicators, the natural mineral biochar of Kerman mountains can inhibit various stages of diseases and help biodiversity.

3

Increase production efficiency

The use of natural mineral biochar from Kerman mountain will increase the content of nutrients in animal and poultry feed, which will lead to higher production efficiency.

Increasing productivity and profitability in the production of livestock and poultry feed using natural mineral biochar from the mountains of Kerman

Reduce costs

The use of mineral biochar from Kohbanan Kerman will lead to a reduction in the production costs of livestock and poultry feed.



Increase efficiency

using _ The natural mineral biochar of Kohbanan Kerman will increase production efficiency and as a result will increase the profitability of the business.





Kouhbanan organic biochar mine .Kerman.

IRAN.

بیوچار طبیعی معدنی کوهبنان کرمان .

ایران .

Eng, Mr.Mohammad Majidi Kouhbanani

آقای مهندس محمد مجیدی کوه بنانی

Methods of using natural mineral biochar from Kerman mountains in the production of animal and poultry feed

1

Direct use

Kerman mountain natural mineral biochar is directly added to livestock and poultry feed as an effective supplement.

2

Combination with feed

biochar normal Mineral Kuhbanan can be used in combination with other ingredients of livestock and poultry feed.

3

earthworks

using _ The natural mineral biochar of Kerman mountains in agricultural lands can lead to the improvement of the quality of fodder and pastures.

Scientific studies and research related to the use of natural mineral biochar from Kerman mountains in the production of animal and poultry feed

Verification of effectiveness

Studies have shown that the use of The natural mineral biochar of Kohbanan Kerman has significantly improved the quality and yield of animal and poultry feed

Natural resources research

Impact success studies The natural mineral biochar of Kerman mountains has been approved for the consumption of natural resources and the environment



The effect of natural mineral biochar of Kerman mountains on egg quality parameters

Eggshell quality

Significant improvement

Egg size and weight

Significant increase

Ovulation factors

Improve responsiveness

Natural mineral biochar Kerman mountains as a suitable alternative for mineral supplements in livestock and poultry diets

جدول ۴- تأثیر دیترهای آزمایشی بر کیفیت پوسته تخمیرغ در دورههای مختلف پرورش												
Table 4- The effect of dietary treatments on egg shell quality in different periods												
تجاربهای آزمایشی Experimental treatments	شاخص شکل تخمیرغ (%) shape index (%)				وزن مخصوص تخمیرغ (گرم بر سانتیمتر مکعب) egg specific gravity (g/cm ³)				مقاومت پوسته تخمیرغ (کنترل گرم بر سانتیمتر مربع) egg breaking strength (kg/cm ²)			
	50-53w	54-57w	58-61w	50-61w	50-53w	54-57w	58-61w	50-61w	50-53w	54-57w	58-61w	50-61w
بیوچار												
Biochar (%RMS) ^۱												
0	80.99	78.35	77.95	79.10	1.085	1.086	1.076	1.083	36.19	32.96	31.15	33.44
25	79.05	78.71	79.01	78.93	1.087	1.093	1.079	1.086	37.15	35.42	33.05	35.15
50	79.30	78.08	79.21	78.87	1.087	1.087	1.079	1.085	38.06	33.15	30.87	34.03
75	78.25	79.40	78.02	78.56	1.083	1.085	1.080	1.082	35.32	30.98	30.18	32.47
100	77.69	78.17	78.36	78.07	1.089	1.088	1.081	1.086	33.99	31.00	29.93	31.94
SEM	0.865	0.687	0.737	0.498	0.002	0.003	0.002	0.001	1.456	1.354	1.598	1.086
P-value	0.097	0.659	0.659	0.626	0.335	0.241	0.329	0.189	0.344	0.139	0.671	0.261
ویتامین C												
Vitamin C (mg/kg)												
0	78.93	78.12	79.23*	78.76	1.087	1.089	1.080	1.085	36.74	33.13	31.19	33.51
100	79.18	78.96	77.79*	78.65	1.085	1.087	1.078	1.083	35.55	32.27	30.88	33.30
SEM	0.547	0.434	0.466	0.315	0.001	0.002	0.001	0.001	0.921	0.857	1.011	0.687
P-value	0.750	0.179	0.034	0.791	0.247	0.521	0.107	0.102	0.369	0.484	0.833	0.826
ویتامین C * بیوچار												
Biochar *Vitamin C												
0×0	82.50	78.98	78.37	79.95	1.084	1.086	1.079	1.083	40.09	34.61	32.33	35.69
0×100	79.49	77.73	77.54	78.25	1.086	1.087	1.073	1.082	32.29	31.31	29.96	31.19
25×0	77.87	77.70	79.17	78.25	1.086	1.091	0.830	1.086	36.12	33.95	34.39	34.71
25×100	80.23	79.72	78.86	79.61	1.089	1.095	1.077	1.087	38.17	36.88	31.71	35.59
50×0	78.53	77.66	79.40	78.53	1.087	1.092	1.080	1.086	38.14	32.68	30.55	33.79
50×100	80.07	78.51	79.02	79.20	1.086	1.083	1.079	1.083	37.99	33.61	31.19	34.26
75×0	77.81	78.58	79.60	78.66	1.087	1.088	1.081	1.085	34.27	32.36	28.94	31.47
75×100	78.69	80.22	76.44	78.45	1.080	1.083	1.079	1.080	36.37	29.60	31.42	33.47
100×0	77.96	77.71	79.64	78.44	1.092	1.088	1.081	1.087	35.05	32.05	29.73	31.90
100×100	77.43	78.64	77.08	77.71	1.085	1.088	1.082	1.085	32.94	29.96	30.14	31.98
SEM	1.223	0.972	1.043	0.704	0.003	0.004	0.002	0.002	2.058	1.909	2.260	1.533
P-value	0.224	0.503	0.530	0.242	0.288	0.393	0.665	0.576	0.109	0.437	0.760	0.324

میانگینهای هر سوسن با حروف غیر مشترک دارای اختلاف معنی داری می باشد (P<0.05).

۱- میانگین شکل تخمیرغ

۳- جدول

میانگین‌های هر ستون با حروف غیر مشابه دارای اختلاف معنی‌دار می‌باشند (P<0.05).
^۱ میانگین شکل تخمیرغ.

جدول ۳- تأثیر داربدهای آزمایشی بر عملکرد مرغهای تخمیرگذار در دورههای مختلف پرورش											
Table 3- The effect of dietary treatments on laying hens performance in different periods											
تجاربهای آزمایشی	مصرف خوراک (گرم/مرغ/روز)				ضریب تبدیل خوراک (گرم مصرفی/گرم تخمیرغ تولیدی)				افزایش وزن روزانه (گرم/مرغ/روز)		
Experimental treatments	Feed intake (g/hen/day)				Feed conversion ratio (g feed/g egg)				Daily weight gain (g/hen/day)		
	50-53w	54-57w	58-61w	50-61w	50-53w	54-57w	58-61w	50-61w	50-53w	58-61w	50-61w
بیوچار											
Biochar (%RMS) ^۱											
0	98.79 *	113.7	114.8	109.1	1.666	1.854	1.847	1.791	12470	13170	0.969
25	99.19 *	115.3	119	111.2	1.656	1.848	1.852	1.788	12720	13460	1.019
50	98.51 *	114.3	114.6	109.1	1.650	1.860	1.816	1.777	12550	13180	0.886
75	96.29 *	112.9	113.2	107.5	1.617	1.826	1.782	1.743	12780	13250	0.652
100	98.96 *	115.7	117.5	110.7	1.682	1.876	1.852	1.806	12790	13340	0.772
SEM	0.617	1.077	2.443	1.215	0.017	0.025	0.043	0.026	0.135	0.127	0.142
P-value	0.013	0.346	0.468	0.232	0.120	0.720	0.727	0.526	0.346	0.471	0.371
ویتامین C											
Vitamin C (mg/kg)											
0	98.85	115	116.3	110.1	1.670	1.875	1.848	1.799	12630	13310	0.945
100	97.84	113.8	115.3	109.0	1.639	1.832	1.811	1.762	12690	13250	0.774
SEM	0.390	0.681	1.545	0.768	0.011	0.016	0.027	0.016	0.086	0.080	0.090
P-value	0.073	0.229	0.631	0.325	0.055	0.064	0.339	0.119	0.624	0.579	0.186
ویتامین C * بیوچار											
Biochar * vitamin C											
0×0	97.72 *	114.9	115.6	109.4	1.683	1.912	1.890	1.830	12390	13200	1.132
0×100	99.86 *	112.6	113.9	108.8	1.650	1.797	1.805	1.752	12550	13130	0.807
25×0	101.41 *	115.4	119.1	112.0	1.699	1.865	1.872	1.814	12940	13690	1.047
25×100	96.98 *	115.2	118.8	110.4	1.614	1.832	1.832	1.762	12510	13220	0.990
50×0	101.57 *	114.2	113.2	109.7	1.703	1.855	1.797	1.786	12450	12980	0.739
50×100	95.45 *	114.3	116	108.6	1.598	1.865	1.834	1.768	12640	13390	1.033
75×0	94.22 *	112.2	112.7	106.4	1.597	1.834	1.795	1.744	12510	13160	0.911
75×100	98.26 *	113.5	113.8	108.6	1.636	1.819	1.771	1.743	13060	13340	0.393
100×0	99.26 *	118	121.1	112.8	1.866	1.906	1.890	1.823	12880	13520	0.897
100×100	98.55 *	113.4	113.8	108.6	1.697	1.846	1.814	1.788	12700	13160	0.647
SEM	0.873	1.523	3.453	1.718	0.025	0.036	0.061	0.037	0.191	0.179	0.204
P-value	0.0001	0.338	0.649	0.481	0.014	0.465	0.863	0.850	0.120	0.108	0.337

میانگین‌های هر ستون با حروف غیر مشابه دارای اختلاف معنی‌دار می‌باشند (P<0.05).
میانگین شکل تخمیرغ.

Easy

The natural mineral biochar of Kerman mountain is easily added to livestock and poultry feed as a mineral supplement.

supplement

At Studies , natural mineral biochar of Kerman mountains as an effective replacement supplement It has been used in feeding livestock and poultry.